Project Charter — Integration Control Center *Version 1.0 | March 2026 | Author: Soumya Singh*

Problem Statement Orders, payments, and customer support data currently exist in separate systems with no unified integration layer. This causes payment reconciliation delays, missed support escalations, and no single view of transaction health. This project builds an end-to-end integration pipeline to connect these systems, automate exception handling, and provide real-time operational visibility.

Systems Involved

- Order Management System — tracks customer orders, items, delivery status, and order lifecycle events
- Billing and Payments System — handles invoice creation, payment processing, reconciliation, and payment failure events
- Customer Support System — manages support ticket creation, prioritization, escalation, and resolution tracking

Business Users

- Operations Team — monitors order fulfillment and delivery SLAs
- Finance Team — tracks payment reconciliation and invoice accuracy
- Support Agents — receive and resolve tickets triggered by payment failures and delivery issues
- Management — views portfolio-level KPIs and exception trends via dashboard

Success Metrics

- 95% or higher payment reconciliation rate across all completed orders
- Support tickets auto-created within 5 minutes of a payment failure event
- Zero silent failures — every exception is logged and traceable
- Dashboard reflects real-time integration health with drill-down to order level
- Full UAT pass rate of 90% or above before go-live sign-off

Final Deliverables

1. Business Requirements Document (BRD)
2. Functional Requirements Document (FRD)
3. Source to Target Mapping
4. SQL Data Model and Validation Pack
5. API Specifications for Orders, Payments, and Support
6. Apache NiFi Integration Flow with Audit Logging
7. Tableau KPI Dashboard
8. UAT Pack, Defect Log, RCA Documents, and Release Notes

Assumptions

- The Olist Brazilian ecommerce dataset will serve as the primary data source
- Support ticket data will be modeled synthetically based on business rules
- All tools used are free or open source versions

Out of Scope

- Live production deployment
- Real payment gateway integration
- Mobile or web front-end development